

MODULE 6 OF 6

Block Models, Cut-off Grade and the Resource Statement

A kriged grid is not yet a resource — this module makes it one, then assesses Toro

This final module turns estimates into tonnes, applies the economics that decide what is ore, attaches a classification, and assembles the formal Resource statement. It closes with a capstone: the honest Toro resource picture set against the Achmmach and Alphamin Bisie benchmarks.

LEARNING OBJECTIVES

- Construct a block model and convert estimated grade and volume into tonnes of ore and metal.
- Apply the volume-variance relationship.
- Derive a breakeven cut-off grade and read a grade-tonnage curve.
- Assign Resource classification to a block model.
- Assemble a code-compliant Resource statement.
- Complete the alluvial g/m³ estimate and benchmark Toro.

1 The block model

A **block model** is the digital orebody: the deposit volume divided into a regular 3D array of rectangular blocks, each carrying estimated attributes (grade of every element, density, classification, rock type, and kriging variance). Block size is a compromise: no smaller than the data can resolve (typically one quarter to one half of the drill spacing) and related to the **selective mining unit** — the smallest volume the method can extract as ore or waste separately. **Sub-blocking** splits edge blocks where a geological boundary cuts through them.

Estimate into blocks, never the reverse

The workflow is strict and one-directional: define domains (Module 3), model the variogram of each (Module 4), then block-krige each domain into its blocks (Module 5). Each block's grade is estimated only from samples in its own domain.

2 From blocks to tonnes

$$T_{\text{block}} = V_{\text{block}} \times \rho, \quad M_{\text{block}} = T_{\text{block}} \times \frac{g}{100} \quad (6.1)$$

The volume-variance relationship

The dispersion variance $D^2(v | D)$ of grades of volumes v within deposit D shrinks as v grows. **Krige's relationship:**

$$D^2(v | D) = D^2(v | G) + D^2(G | D) \quad (6.2)$$

The variance of small blocks within a deposit equals their variance within a panel G plus the variance of panels within the deposit. Variance partitions cleanly across scales. Selective mining units have a wider grade distribution than large panels; a model built on the wrong support will mis-state how much ore sits above cut-off. It is the formal reason kriging smooths — a kriged block model cannot reproduce the full block-scale variance, which is what conditional simulation was created to fill.

3 Cut-off grade

The **cut-off grade** divides material worth processing from material that is not. It is fundamentally an economic quantity. The breakeven (marginal) cut-off is where revenue from treating a tonne as ore equals processing cost. A tonne at grade $g\%$ contains $g/100$ tonnes of metal:

$$\frac{g_c}{100} \times P \times R = C \quad \implies \quad g_c = \frac{100 C}{P \times R} \quad (6.3)$$

A Resource is always reported **at a stated cut-off**, resting on realistic, disclosed assumptions for price P , recovery R and cost C .

The grade-tonnage curve

Sweep the cut-off and plot the surviving tonnage and the mean grade of that tonnage — the most informative graphic a resource geologist produces.

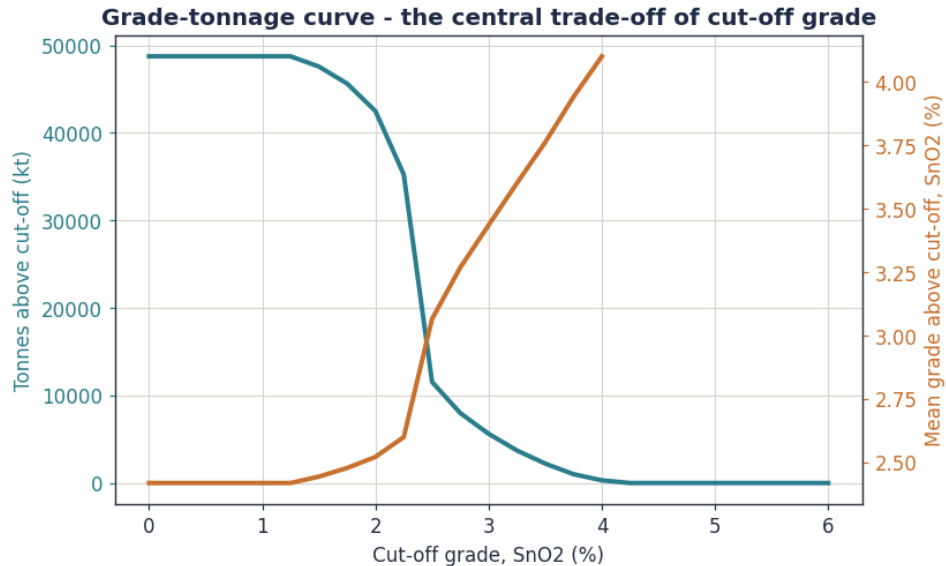


Figure 1 – Toro grade-tonnage relationship. As cut-off rises, tonnage falls and mean grade rises. Contained metal — the product — usually peaks at a low cut-off and then falls. The “right” cut-off maximises project value, not grade.

4 Classification in practice

Module 1 defined the categories; here we assign them to blocks. Evidence: **drill / pit spacing** (the primary driver), **kriging variance** (a direct block-by-block confidence map), the **number and configuration of informing samples**, **conditional simulation** (used to test whether grade in a production-sized volume is known to a specified tolerance), and **geological continuity plus QAQC**.

Classifying purely on a raw kriging-variance threshold scatters isolated Measured blocks through an Indicated background — unmineable. The CP defines confidence as broad, coherent regions, commonly via drill-spacing criteria validated by conditional simulation, not block by isolated block.

5 The Mineral Resource statement

Classification	Tonnes	Grade	Contained metal
Measured	by domain	% Sn / g/m ³	tonnes Sn
Indicated	by domain	...	...
Measured + Indicated	subtotal	...	...
Inferred	reported separately	...	...

Measured and Indicated are subtotalled because only they can feed a Reserve; Inferred is always kept separate. The table never travels alone: it must carry the effective date, cut-off grade and price / recovery / cost assumptions, the RPEEE basis, and the named CP sign-off.

Rounding is a disclosure

An Inferred Resource quoted as “5,234,718 t at 1.07% Sn” is dishonest — the false precision contradicts the category. “5.2 Mt at 1.1% Sn” tells the truth.

6 Conditional simulation, briefly

Kriging gives the single best estimate and, by smoothing, deliberately understates true block-grade variability. **Conditional simulation** generates many alternative block models — *realisations* — each of which honours the data and reproduces the real histogram and variogram. The spread across realisations *is* the uncertainty.

Use **kriging** for the best single estimate — the resource block model itself. Use **conditional simulation** for uncertainty: risk analysis, classification criteria, grade-control decisions, recoverable resources. Complementary, not competing.

7 Reconciliation and the path to a Reserve

Reconciliation compares the resource model, the grade-control model and what the processing plant actually received — usually expressed as mine-call factors. Persistent divergence almost always traces to a sampling, density, domaining or top-cut problem. Beyond the Resource lies the Reserve: Measured and Indicated blocks pass into Scoping, Pre-Feasibility and Feasibility studies that apply the modifying factors (mining method, dilution, recovery, costs, permits, markets). That is the journey Achmmach completed to publish 7.0 Mt at 0.82% Sn Proven and Probable Reserve.

8 Completing the alluvial estimate

The block-model machinery above was demonstrated on the kriged concentrate grade. A defensible *alluvial* resource is a volume-grade quantity in g/m³, with the XRF assay supplying only one factor in equation (2.4):

$$g_{\text{Sn}} \left[\frac{\text{g}}{\text{m}^3} \right] = \frac{M_{\text{HMC}}}{M_{\text{bulk}}} \times a_{\text{Sn}} \times \rho_{\text{gravel}} \times 10^6 \quad (6.4)$$

Applied to the Toro field recoveries (200 kg bulk samples), with mean concentrate Sn grade $\approx 1.8\%$ and assumed gravel density 1.9 t/m³:

Bulk sample	HMC yield	In-situ Sn (g/m ³)
Mining Pit (pt 2)	1.00%	339
River Bishwai (pt 13)	1.00%	339
Kadade (pt 11)	0.50%	169
Kadade (pt 16)	0.50%	169
GN-005	0.18%	59

Mean ≈ 215 g/m³ Sn, range 59–339 g/m³ — an order-of- magnitude span from just five samples.

Equation (6.4) is a *product*, so the relative errors of its factors add. A 20% error in yield, 15% in concentrate grade and 10% in density compound to roughly 27% in g/m³. The 215 g/m³ mean rests on five bulk samples, an *assumed* density that has never been measured at Toro, and a yield read from field washing that had no QAQC. The pitting programme must deliver, for every pit, a weighed bulk sample of measured volume, a measured in-situ density, and a logged wash-layer thickness. Without those, the alluvial estimate cannot rise above an Exploration Target.

9 Capstone: the Toro resource picture

Status. Two distinct Exploration Targets — an alluvial placer across Bishiwai, Yadabongo and Kadade, and a primary greisen-pegmatite target on the LV 029 cupola. Neither is a Mineral Resource, and the reasons are now precise.

Why not yet a Resource. The grade data resolve to 26 pit locations (Module 2). The variogram has a relative nugget of 0.55 (Module 4). Cross-validation returns correlation 0.24: unbiased but imprecise (Module 5). The alluvial g/m³ figure rests on five bulk samples and an unmeasured density (Module 6).

What the data do support. A real, encouraging signal: structured grade continuity of roughly 1 km, concentrate grades averaging 2.3% SnO₂, a reconstructed alluvial grade near 215 g/m³ Sn, and a coherent higher-grade core through Bishiwai-Yadabongo. Enough to justify the next tranche of capital — which is what an Exploration Target is for.

The path from Exploration Target to Mineral Resource

Action	Modules	What it changes
Drill the 13 priority collars; systematic infill pitting	2, 5	Raises density so continuity can be assumed, not implied
Designed QAQC stream from the first sample	2	Lowers the nugget; makes the database certifiable
Measure in-situ bulk density by domain	2, 6	Removes the largest assumption in tonnage and the g/m ³ chain
Record wash-layer thickness on every pit	3, 6	Enables correct length-weighted compositing
Re-fit variograms per domain; block-krige; classify with QKNA and simulation	4–6	Converts the Exploration Target into a classified Resource
CP sign-off and a code-compliant report	1, 6	Makes the Resource public and bankable

10 The workflow as one chain

The resource estimation workflow

1. Foundations — know what you are producing, under which code, and who is accountable. **2. Sampling and data** — collect representative samples, control the error, certify the database. **3. EDA and statistics** — understand the distribution and define the stationary domains. **4. Variography** — measure and model the spatial continuity of grade. **5. Estimation** — kriging the best linear unbiased estimate, and obtain its quantified uncertainty. **6. Reporting** — turn estimates into classified tonnes and metal, honestly. Each module is necessary; none is sufficient alone.

Hold on to the deeper lesson, the one the Toro data taught more forcefully than any clean textbook dataset could. Resource estimation is not the manufacture of a single confident number. It is the disciplined, quantified, defensible *statement of what is known and how well it is known* — including, when the data demand it, the honest verdict that the answer is “not yet, and here is precisely why.” A geologist who can deliver that verdict with the rigour of this course, and chart

the path beyond it, is doing resource estimation at the highest level.

Module 6 closes the Toro Resource Estimation & Geostatistics course.